

Yue Xue (薛岳)

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<https://github.com/BradMoonUESTC>

<https://scholar.google.com/citations?user=6cdLQQQAAAAJ>

SUMMARY

- Senior Security Engineer focused on blockchain security, software security, and LLM applications in security.
- An early LLM4Sec researcher and builder: co-author of foundational, highly-cited works (GPTScan, PropertyGPT), advancing a security methodology combining LLMs with program analysis/formal verification, and building **Finite Monkey** to industrialize AI smart contract auditing at scale.
- Owned internal anti-money laundering (AML) initiatives at **OKX** with end-to-end on-chain compliance experience, from fund monitoring and risk identification to alerting/interdiction and closed-loop handling.
- Experienced in smart contract auditing, static analysis, formal verification, and on/off-chain monitoring systems.
- Awards: **ETHSecurity Badge** holder; **NDSS 2025 Distinguished Paper Award**; **ACM SIGSOFT Distinguished Paper Award (FSE 2024)**; **Secure3 Rank #2** (total bounty USD 68,570).

EDUCATION

Guangzhou University (GZHU) 2020.09 – 2023.06

M.S. in Computer Technology (Academician Fang Binxing Class)

- Research: blockchain security, smart contract security, LLM for security

University of Electronic Science and Technology of China (UESTC) 2011.09 – 2015.06

B.E. in Electrical Engineering and Automation

- Focus: electrical engineering, automatic control, and core engineering coursework

EXPERIENCE

CertiK 2025.11 – Present

Head of AI Research

- Lead developer for **CertiK** AI Auditor, owning core engineering and new auditing methods.
- Own the development and optimization of finding deduplication and validation modules.
- Refine validation quality using auditor review and competition judging criteria to improve finding accuracy.
- Own core implementation across code analysis, candidate discovery, finding validation, and reporting; translate in-house auditing methods into engineering delivery.
- Build evaluation and regression workflows, compare findings with reference results, and investigate failure cases to improve detection quality.
- Responsible for building self-evolving workflows.
- Improve execution reliability and computational efficiency, and integrate research capabilities into practical auditing workflows.

OKX 2022.04 – 2022.10, 2024.05 – 2025.11

Senior Security Engineer

- Owned the development and operation of ATMS (Asset Transaction Monitoring System), responsible for first-line AML risk control covering external deposits into **OKX**.
- Led high-risk address tracing, linked-address expansion, address intelligence maintenance, and risk assessment, supported by manual on-chain fund-flow investigations.
- Owned risk statistics, analysis, and reporting across monitoring and investigation workflows to support day-to-day risk-control operations.
- Led a 4-person team to build internal contract security monitoring (state and transaction anomaly detection)
- Designed and maintained the internal contract management platform and privileged-task framework; delivered safe privilege migration with zero incidents
- Built AML monitoring for CeFi deposits and wallet transactions; continuously improved black-fund detection and interception
- Conducted smart contract auditing and security tooling development

MetaTrust 2022.10 – 2025.08

Security Supervisor / Consultant

- Led smart contract audits and engagement coordination; improved audit workflows
- Developed 60+ **Slither** detectors and optimized nearly 100 rules; reduced false positives below 25%
- Contributed to GPTScan and independently built PropertyGPT for formal verification
- Collaborated with **NTU** research teams and co-authored papers

C# Developer 2016.12 – 2019.01

- Built full-stack web features and maintained databases
- Developed and maintained enterprise systems

LLM4Sec Audit Tools

Ongoing

End-to-end audit toolchain

- **Code Audit Hinder** (IDE security plugin, 39 stars, 2025.03–Present): RAG-powered audit hints for VSCode and Cursor
- **Finite Monkey Engine** (360 stars, 2024.04–Present): task-driven LLM audit engine with practical precision/recall
- **TxAnalyzer** (GitHub, 2026.05–Present): agent skill and CLI for blockchain attack transaction root-cause analysis, with EVM/Solana artifact collection
- AI Audit Result Evaluator (private, 2025.05–Present): automated judging for contest findings vs. ground truth
- Code Semantic Search (private, 2024.12–Present): LLM-based codebase understanding and retrieval

Falcon

2022.12 – 2024.05

Enhanced Solidity static analysis framework

- Extended **Slither** with dozens of new detectors and improved precision
- Supports DeFi price manipulation, centralization risk, and complex logic bug detection

Auditing & Competitions

Ongoing

Secure3 track

- **Secure3 Rank #2**, total bounty USD 68,570

PUBLICATIONS

1. Y. Sun, D. Wu, **Yue Xue**, H. Liu, H. Wang, Z. Xu, X. Xie, Y. Liu. “GPTScan: Detecting Logic Vulnerabilities in Smart Contracts by Combining GPT with Program Analysis.” Proc. ICSE, 2024.
2. Y. Sun, D. Wu, **Yue Xue**, H. Liu, W. Ma, L. Zhang, Y. Liu, Y. Li. “LLM4Vuln: A Unified Evaluation Framework for Decoupling and Enhancing LLMs’ Vulnerability Reasoning.” arXiv preprint arXiv:2401.16185, 2024.
3. W. Ma, D. Wu, Y. Sun, T. Wang, S. Liu, J. Zhang, **Yue Xue**, Y. Liu. “Combining Fine-Tuning and LLM-based Agents for Intuitive Smart Contract Auditing with Justifications.” Proc. ICSE, 2025.
4. Y. Liu, **Yue Xue**^{*}, D. Wu, Y. Sun, Y. Li, M. Shi, Y. Liu. “PropertyGPT: LLM-driven Formal Verification of Smart Contracts through Retrieval-Augmented Property Generation.” NDSS, 2025.
5. K. Li, **Yue Xue**, S. Chen, H. Liu, K. Sun, M. Hu, H. Wang, Y. Liu, Y. Chen. “Static Application Security Testing (SAST) Tools for Smart Contracts: How Far Are We?” Proc. ACM Softw. Eng. (FSE), 1, 1447–1470, 2024.
6. **Yue Xue**, J. Fu, S. Su, Z. A. Bhuiyan, J. Qiu, H. Lu, N. Hu, Z. Tian. “Preventing Price Manipulation Attack by Front-Running.” Proc. AAIS, 309–322, 2022.
7. M. Xie, M. Hu, Z. Kong, C. Zhang, Y. Feng, H. Wang, **Yue Xue**, H. Zhang, Y. Liu, et al. “DeFort: Automatic Detection and Analysis of Price Manipulation Attacks in DeFi Applications.” Proc. ISSTA, 2024.
8. **Yue Xue**, D. Fan, S. Su, J. Fu, N. Hu, W. Liu, Z. Tian. “A Review on the Security of the Ethereum-Based DeFi Ecosystem.” CMES—Computer Modeling in Engineering & Sciences, 139(1), 2024.
9. S. Su, Y. Tan, **Yue Xue**, C. Wang, H. Lu, Z. Tian, C. Shan, X. Du. “Detecting Smart Contract Project Anomalies in Metaverse.” Proc. IEEE MetaCom, 2023.
10. Z. Kong, C. Zhang, M. Xie, M. Hu, **Yue Xue**, Y. Liu, H. Wang, Y. Liu. “Smart Contract Fuzzing Towards Profitable Vulnerabilities.” Proc. ACM Softw. Eng. (FSE), 2, 153–175, 2025.
11. S. Su, **Yue Xue**, L. Lin, C. Wang, H. Lu, J. Qiu, Y. Sun, Y. Liu, Z. Tian. “Smart Contract Firewall: Protecting the On-Chain Smart Contract Projects.” Proc. IEEE GLOBECOM, 5122–5128, 2023.
12. Y. Sun, S. Ding, L. Tan, **Yue Xue**, B. Li, C. Huang, S. Xiong. “Beyond Stars and Commits: A Fair and Manipulation-Resistant Framework for Measuring Developer Contributions.” Proc. ACM FSE Companion, 2025.
13. J. Gao, **Yue Xue**, X. Xie, J. Cao, S. Thant, E. Lee, B. Xu. “Understanding Codebase like a Professional! Human–AI Collaboration for Code Comprehension.” ICPC ’26.
14. **Yue Xue**. “When and How Context Rot Appears in Coding Agents: A White-Box Study of Agent Skills in Code Auditing.” arXiv ’26 (preprint).
15. **Yue Xue**. “Chiral Analysis of Smart Contracts: Detecting Vulnerabilities from Relational Inconsistencies Across Business Paths.” arXiv ’26 (preprint).
16. Z. Kong, W. Xia, C. Wang, **Yue Xue**, Y. Lu, P. Li, S. Li, Z. Cao, Y. Liu. “Knowdit: Agentic Smart Contract Vulnerability Detection with Auditing Knowledge Summarization.” arXiv ’26 (preprint).
17. R. Liang, J. Chen, X. Li, H. Gu, Y. Feng, **Yue Xue**, C. Wu, Y. Liu. “EvoPoC: Automated Exploit Synthesis for DeFi Smart Contracts via Hierarchical Knowledge Graphs.” arXiv ’26 (preprint).

* Co-first author.

ACADEMIC SERVICE

- Reviewer: [IEEE Transactions on Software Engineering \(TSE\)](#); [ACM Transactions on Software Engineering and Methodology \(TOSEM\)](#); [ACM Transactions on Privacy and Security \(TOPS\)](#)
- Reviewer: [1st Workshop on Interpreting Agent Behavior \(IAB\)](#), NeurIPS 2026.

INDUSTRY WRITING

- Authored widely circulated security and AI research articles, including X posts with tens of thousands of views: [article 1](#); [article 2](#).
- Author: [AI 审计方法论：你需要知道的一切（第一部分）](#) , Chinese edition on [登链社区](#), March 16, 2026.
- Author: [AI 安全审计智能体核心方法论（第二部分）](#) , Chinese edition on [登链社区](#), April 1, 2026.

MEDIA INTERVIEW

- Interviewed expert: [International Financial News](#), August 18, 2026, [接连“越界”，AI 失控的警报已经拉响？](#) , discussing AI agent safety boundaries and safeguards.

AWARDS

- [NDSS 2025 Distinguished Paper Award](#): [PropertyGPT](#)
- [ACM SIGSOFT Distinguished Paper Award \(FSE 2024\)](#): [SAST Tools for Smart Contracts](#)
- [GLOBECOM 2023 Best Paper Award](#): [Smart Contract Firewall](#)
- [ETHSecurity Badge](#) holder
- [Secure3 Rank #2](#) (total bounty USD 68,570)